

7	(a)		$F = (\Delta v) \frac{Nm}{t} = (m\Delta v) \frac{N}{t} = 1.85 \times 10^{-23} \times 1.49 \times 10^{24}$ $= 27.6 \text{ N}$	
	(b)		$p = \frac{F}{A} = \frac{27.6}{0.018} = 1530 \text{ Pa} \quad (\text{accept } p = \frac{27.6}{(0.134)^2})$	
	(c)		From $pV = NkT$ $N = \frac{pV}{kT}$ $N = \frac{(1530)(0.018 \times 0.134)}{(1.38 \times 10^{-23})(273+27)}$ $= 8.9 \times 10^{20}$	
	(d)		From $\frac{1}{2}mc_{rms}^2 = \frac{3}{2}kT$ $c_{rms} = \sqrt{\frac{3kT}{m}}$ $c_{rms} = \sqrt{\frac{3(1.38 \times 10^{-23})(273+27)}{6.86 \times 10^{-27}}}$ $= 1350 \text{ ms}^{-1}$ Accept $\Delta p = 2mc_{rms}$ although this is not the preferred method unless student mentioned that based on the assumptions of kinetic theory of gases, the collision between the gas molecules and the container is perfectly elastic.	

8	(a)	(i)	$R = \rho l / A$	
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